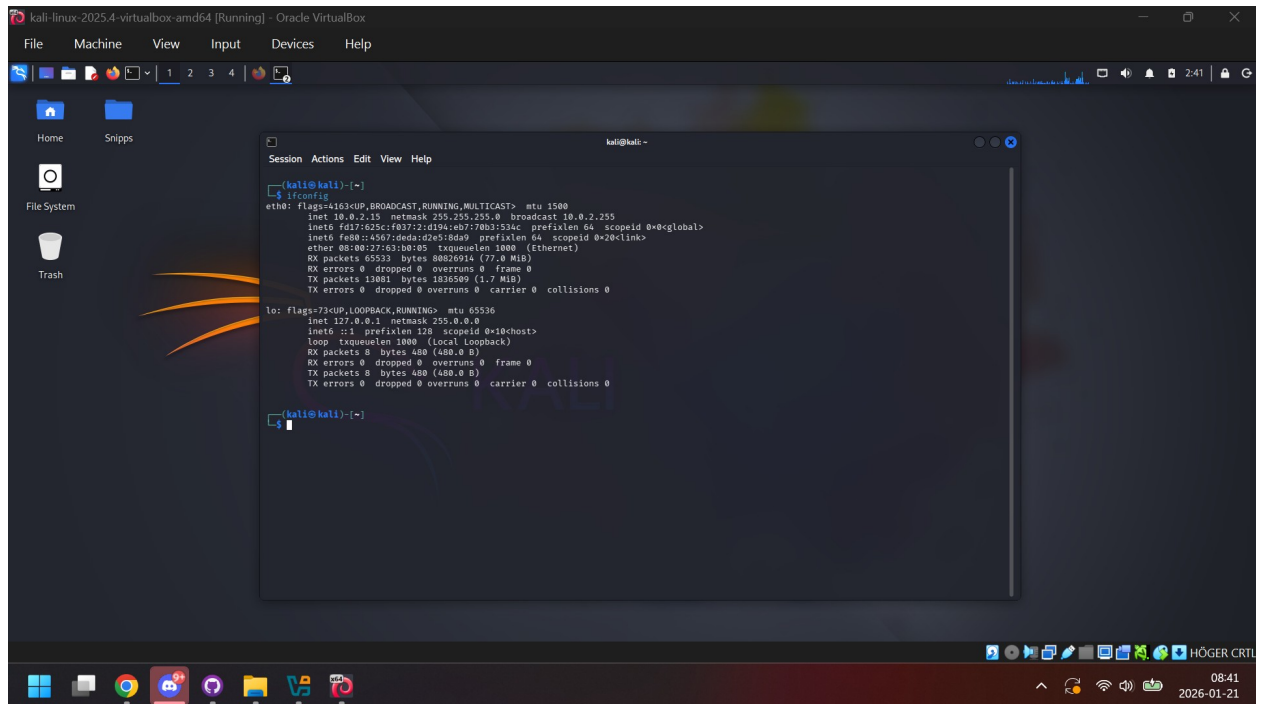


Ethical Hacking - Lab 1

My instance. Kali Linux in Virtualbox:



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

(kali@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.0 netmask 255.0 broadcast 10.0.2.255
    inet6 fd17:025c:f037:2:d194:e07:70b3:534c prefixlen 64 scopeid 0<global>
    inet6 fe80::4567:dada:d2e5:8da9 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:63:b0:a5 txqueuelen 1000 (Ethernet)
    RX packets 65533 bytes 80826914 (77.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13001 bytes 1836500 (1.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)~$
```

1.1

a) USB media storage can be found in the "media" directory:

```
kali@kali: /  
Session Actions Edit View Help  
  
(kali@kali)-[/home]  
$ ls  
kali  
  
(kali@kali)-[/home]  
$ cd ..  
  
(kali@kali)-[/]  
$ ll  
total 976632  
lrwxrwxrwx 1 root root 7 Nov 10 04:50 bin → usr/bin  
drwxr-xr-x 3 root root 4096 Dec 2 22:03 boot  
drwxr-xr-x 18 root root 3240 Jan 20 11:05 dev  
drwxr-xr-x 180 root root 12288 Jan 20 11:05 etc  
drwxr-xr-x 3 root root 4096 Dec 2 21:36 home  
lrwxrwxrwx 1 root root 33 Dec 2 22:02 initrd.img → boot/initrd.img-6.16.8+kali-amd64  
lrwxrwxrwx 1 root root 33 Dec 2 22:02 initrd.img.old → boot/initrd.img-6.16.8+kali-amd64  
lrwxrwxrwx 1 root root 7 Nov 10 04:50 lib → usr/lib  
lrwxrwxrwx 1 root root 9 Dec 2 21:33 lib32 → usr/lib32  
lrwxrwxrwx 1 root root 9 Nov 10 04:50 lib64 → usr/lib64  
drwx----- 2 root root 16384 Dec 2 22:01 lost+found  
drwxr-xr-x 2 root root 4096 Dec 2 21:29 media  
drwxr-xr-x 3 root root 4096 Jan 20 11:09 mnt  
drwxr-xr-x 3 root root 4096 Dec 2 21:33 opt  
dr-xr-xr-x 251 root root 0 Jan 20 11:05 proc  
drwx----- 5 root root 4096 Jan 20 12:14 root  
drwxr-xr-x 36 root root 860 Jan 20 11:05 run  
lrwxrwxrwx 1 root root 8 Nov 10 04:50 sbin → usr/sbin  
drwxr-xr-x 3 root root 4096 Dec 2 21:34 srv  
-rw----- 1 root root 1000000000 Dec 2 22:02 swap  
dr-xr-xr-x 13 root root 0 Jan 20 12:02 sys  
drwxrwxrwt 13 root root 340 Jan 21 02:39 tmp  
drwxr-xr-x 15 root root 4096 Dec 2 21:33 usr  
drwxr-xr-x 12 root root 4096 Jan 19 10:53 var  
lrwxrwxrwx 1 root root 30 Dec 2 22:02 vmlinuz → boot/vmlinuz-6.16.8+kali-amd64  
lrwxrwxrwx 1 root root 30 Dec 2 22:02 vmlinuz.old → boot/vmlinuz-6.16.8+kali-amd64
```

b) I ran the command "sudo systemctl start NetworkManagement". The mount has already been done ones before, thats why its "busy":

```
kali@kali: /mnt/share  
Session Actions Edit View Help  
  
(kali@kali)-[/]  
$ sudo mount -t cifs //192.168.0.193/Shared /mnt/share \  
-o vers=3.0,username=,password, rw  
mount error(16): Device or resource busy  
Refer to the mount.cifs(8) manual page (e.g. man mount.cifs) and kernel log messages (dmesg)  
  
(kali@kali)-[/]  
$ cd /mnt/share/  
  
(kali@kali)-[/mnt/share]  
$ ll  
total 169  
-rwxr-xr-x 1 root root 19 Jan 20 12:14 Hello.txt  
-rwxr-xr-x 1 root root 169727 Jan 20 12:27 Screenshot_2026-01-20_12-27-34.png  
  
(kali@kali)-[/mnt/share]  
$
```

c) In the images below super user processes. These runs as root and starts at boot.

```
kali@kali: /
Session Actions Edit View Help

(kali@kali)-[/]
$ ps -eo user,cmd | grep "/sbin"
root    /sbin/init splash
root    /usr/sbin/haveged --Foreground --verbose=1
root    /usr/sbin/NetworkManager --no-daemon
root    /usr/sbin/ModemManager
root    /usr/sbin/cron -f
root    /usr/sbin/VBoxService
root    /usr/sbin/lightdm
root    /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d - linux
pcscd   /usr/sbin/pcscd --foreground --auto-exit
kali    grep --color=auto /sbin

(kali@kali)-[/]
$ ps -eo user,cmd | grep "/sbin" | grep root
root    /sbin/init splash
root    /usr/sbin/haveged --Foreground --verbose=1
root    /usr/sbin/NetworkManager --no-daemon
root    /usr/sbin/ModemManager
root    /usr/sbin/cron -f
root    /usr/sbin/VBoxService
root    /usr/sbin/lightdm
root    /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d - linux

(kali@kali)-[/]
$
```

d) There are no active listening TCP or UDP ports. This is due to no network services are enabled.

```
kali@kali: /
Session Actions Edit View Help

(kali@kali)-[~/mnt]
$ sudo sshfs -o allow_other,default_permissions,idmap=user,uid=$(id -u),gid=$(id -g) \
h23samum@maggie.du.se:/home/student/h23samum /mnt/remote_data
h23samum@maggie.du.se's password:

(kali@kali)-[~/mnt]
$ ls -ld /mnt/remote_data
drwx----- 1 kali kali 4096 Jan 23 06:03 /mnt/remote_data

(kali@kali)-[~/mnt]
$ cd /mnt/remote_data

(kali@kali)-[/mnt/remote_data]
$ ll
total 20
drwx----- 1 kali kali 4096 Mar 18 2024 Documents
drwx----- 1 kali kali 4096 Mar 18 2024 Downloads
drwx----- 1 kali kali 4096 Mar 18 2024 Favorites
drwx----- 1 kali kali 4096 Mar 18 2024 'Saved Games'
drwx----- 1 kali kali 4096 Mar 18 2024 Windesk

(kali@kali)-[/mnt/remote_data]
$ cd Saved\ Games

(kali@kali)-[/mnt/remote_data/Saved Games]
$ ll
total 8
drwx----- 1 kali kali 4096 Mar 18 2024 '$RECYCLE.BIN'
-rwx----- 1 kali kali 282 Mar 18 2024 desktop.ini
```

a)

```
kali@kali: /  
Session Actions Edit View Help  
  
(kali@kali)-[~/mnt]  
$ sudo sshfs -o allow_other,default_permissions,idmap=user,uid=$(id -u),gid=$(id -g) \  
h23samum@maggie.du.se:/home/student/h23samum /mnt/remote_data  
h23samum@maggie.du.se's password:  
  
(kali@kali)-[~/mnt]  
$ ls -ld /mnt/remote_data  
drwx----- 1 kali kali 4096 Jan 23 06:03 /mnt/remote_data  
  
(kali@kali)-[~/mnt]  
$ cd /mnt/remote_data  
  
(kali@kali)-[/mnt/remote_data]  
$ ll  
total 20  
drwx----- 1 kali kali 4096 Mar 18 2024 Documents  
drwx----- 1 kali kali 4096 Mar 18 2024 Downloads  
drwx----- 1 kali kali 4096 Mar 18 2024 Favorites  
drwx----- 1 kali kali 4096 Mar 18 2024 'Saved Games'  
drwx----- 1 kali kali 4096 Mar 18 2024 Windesk  
  
(kali@kali)-[/mnt/remote_data]  
$ cd Saved\ Games  
  
(kali@kali)-[/mnt/remote_data/Saved Games]  
$ ll  
total 8  
drwx----- 1 kali kali 4096 Mar 18 2024 '$RECYCLE.BIN'  
-rwx----- 1 kali kali 282 Mar 18 2024 desktop.ini
```

b)

```
kali@kali: /
Session Actions Edit View Help

(kali@kali)-[/]
$ cat /proc/cpuinfo
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 166
model name     : Intel(R) Core(TM) i7-10710U CPU @ 1.10GHz
stepping       : 0
microcode      : 0xffffffff
cpu MHz        : 1608.005
cache size     : 12288 KB
physical id    : 0
siblings       : 2
core id        : 0
cpu cores      : 2
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 22
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm co
nstant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes xsave avx
f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ibrs_enhanced fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_clear flush_l1d
arch_capabilities
bugs           : spectre_v1 spectre_v2 spec_store_bypass swapgs mmio_stale_data retbleed eibrs_pbrsb bhi spectre_v2_user its
bogomips       : 3216.01
clflush size   : 64
cache_alignm   : 64
address sizes  : 39 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id      : GenuineIntel
cpu family     : 6
model          : 166
model name     : Intel(R) Core(TM) i7-10710U CPU @ 1.10GHz
stepping       : 0
microcode      : 0xffffffff
cpu MHz        : 1608.005
cache size     : 12288 KB
physical id    : 0
siblings       : 2
core id        : 1
cpu cores      : 2
apicid         : 1
initial apicid : 1
fpu            : yes
fpu_exception  : yes
cpuid level    : 22
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm co
nstant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes xsave avx
f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ibrs_enhanced fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_clear flush_l1d
arch_capabilities
bugs           : spectre_v1 spectre_v2 spec_store_bypass swapgs mmio_stale_data retbleed eibrs_pbrsb bhi spectre_v2_user its
bogomips       : 3216.01
clflush size   : 64
cache_alignm   : 64
address sizes  : 39 bits physical, 48 bits virtual
power management:

(kali@kali)-[/]
$
```

c) journalctl -b shows the full boot log.

```
kali@kali: /
Session Actions Edit View Help

(kali@kali)-[/]
$ journalctl -b
Jan 20 11:05:11 kali kernel: Linux version 6.16.8+kali-amd64 (dev@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.3.0-8) 14.3.0, GNU ld (GNU >
Jan 20 11:05:11 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.16.8+kali-amd64 root=UUID=8a84b405-db00-4b28-a5bc-48ecaa65880d ro quiet>
Jan 20 11:05:11 kali kernel: BIOS-provided physical RAM map:
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000009fbfff] usable
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x0000000000009fc00-0x0000000000009ffff] reserved
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x000000000000f0000-0x000000000000ffffff] reserved
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x00000000000100000-0x0000000000007ffff] usable
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x000000000007fff0000-0x0000000000007ffff] ACPI data
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Jan 20 11:05:11 kali kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffff] reserved
Jan 20 11:05:11 kali kernel: NX (Execute Disable) protection: active
Jan 20 11:05:11 kali kernel: APIC: Static calls initialized
Jan 20 11:05:11 kali kernel: SMBIOS 2.5 present.
Jan 20 11:05:11 kali kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Jan 20 11:05:11 kali kernel: DMI: Memory Slots populated: 0/0
Jan 20 11:05:11 kali kernel: Hypervisor detected: KVM
Jan 20 11:05:11 kali kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Jan 20 11:05:11 kali kernel: kvm-clock: using sched offset of 19323569576 cycles
Jan 20 11:05:11 kali kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns
Jan 20 11:05:11 kali kernel: tsc: Detected 1608.005 MHz processor
Jan 20 11:05:11 kali kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Jan 20 11:05:11 kali kernel: e820: remove [mem 0x000a0000-0x000ffff] usable
Jan 20 11:05:11 kali kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
Jan 20 11:05:11 kali kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
Jan 20 11:05:11 kali kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Jan 20 11:05:11 kali kernel: CPU MTRRs all blank - virtualized system.
Jan 20 11:05:11 kali kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Jan 20 11:05:11 kali kernel: RAMDISK: [mem 0x1f5c7000-0x2badaff]
Jan 20 11:05:11 kali kernel: ACPI: Early table checksum verification disabled
Jan 20 11:05:11 kali kernel: ACPI: RSDP 0x000000000000E000 000024 (v02 VBOX )
Jan 20 11:05:11 kali kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
Jan 20 11:05:11 kali kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
Jan 20 11:05:11 kali kernel: ACPI: DSDT 0x000000007FFF02F0 002353 (v02 VBOX VBOXBIOS 00000002 VBOX 00029BF3)
Jan 20 11:05:11 kali kernel: ACPI: FACS 0x000000007FFF0200 000040
Jan 20 11:05:11 kali kernel: ACPI: FACS 0x000000007FFF0200 000040
Jan 20 11:05:11 kali kernel: ACPI: APIC 0x000000007FFF0240 00005C (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
Jan 20 11:05:11 kali kernel: ACPI: SSDT 0x000000007FFF02A0 000045 (v01 VBOX VBOXCPUPT 00000002 VBOX 00029BF3)
Jan 20 11:05:11 kali kernel: ACPI: Reserving FACP table memory at [mem 0x7fff00f0-0x7fff01e3]
```

d) The autoheader and aclocal-1.15 was missing. I installed them through by using the command "sudo apt -y build essential autoconf automake libncurses-dev" which resolved the problem.

There after i could run "./config" and "make" in the mtr directory to compile the software.

```
kali@kali: /usr/src/mtr-0.92
Session Actions Edit View Help
config.status: creating config.h
config.status: config.h is unchanged
config.status: executing depfiles commands

(kali@kali)-[/usr/src/mtr-0.92]
$ make
(CDPATH="${ZSH_VERSION+.}:" && cd . && /bin/bash /usr/src/mtr-0.92/build-aux/missing autoheader)
aclocal.m4:17: warning: this file was generated for autoconf 2.69.
You have another version of autoconf. It may work, but is not guaranteed to.
If you have problems, you may need to regenerate the build system entirely.
To do so, use the procedure documented by the package, typically 'autoreconf'.
rm -f stamp-h1
touch config.h.in
cd . && /bin/bash ./config.status config.h
config.status: creating config.h
make all-am
make[1]: Entering directory '/usr/src/mtr-0.92'
CC      ui/mtr-mtr.o
ui/mtr.c: In function 'init_fld_options':
ui/mtr.c:298:5: warning: 'memset' used with length equal to number of elements without multiplication by element size [-Wmemset-elt-size]
298 |     memset(ctl->fld_index, -1, FLD_INDEX_SZ);
    |     ^~~~~~
CC      ui/mtr-net.o
CC      ui/mtr-cmdpipe.o
CC      ui/mtr-dns.o
CC      ui/mtr-raw.o
CC      ui/mtr-split.o
ui/split.c: In function 'split_redraw':
ui/split.c:105:52: warning: '%d' directive output may be truncated writing between 1 and 11 bytes into a region of size between 0 and 255 [-Wformat-truncation=]
105 |         snprintf(newLine, sizeof(newLine), "%s %d %d %d %d %d",
    |         ^~~~~~
ui/split.c:105:48: note: directive argument in the range [-2147483, 2147483]
105 |         snprintf(newLine, sizeof(newLine), "%s %d %d %d %d %d",
    |         ^~~~~~
ui/split.c:105:48: note: directive argument in the range [-2147483, 2147483]
ui/split.c:105:52: note: directive argument in the range [-2147483, 2147483]
ui/split.c:105:13: note: 'snprintf' output between 13 and 319 bytes into a destination of size 256
105 |         snprintf(newLine, sizeof(newLine), "%s %d %d %d %d %d",
    |         ^~~~~~
106 |         name, net_loss(at), net_returned(at), net_xmit(at),
```

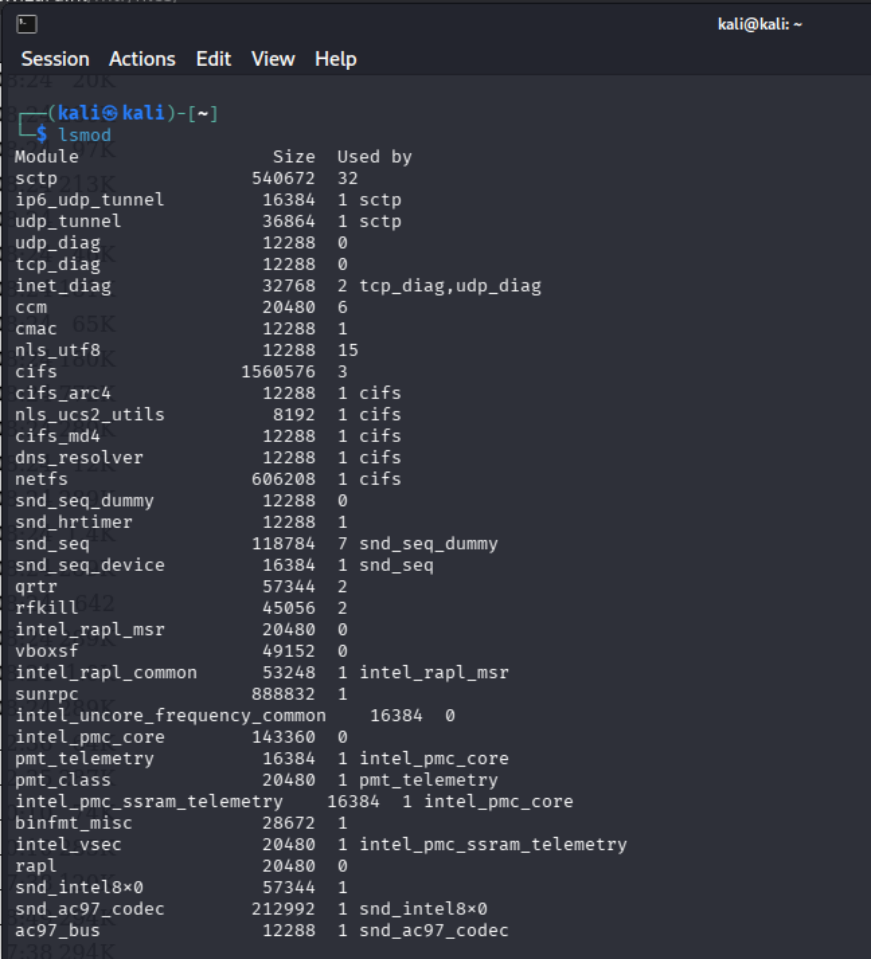
e) I downloaded the 0.94.diff and used the command "patch -p1 < mtr-0.94.iff". It gave some

```
kali@kali: /usr/s
Session  Actions  Edit  View  Help
patching file ui/mtr.c
Hunk #1 succeeded at 295 (offset -4 lines).
Hunk #2 succeeded at 344 (offset -5 lines).
Hunk #3 succeeded at 441 (offset -6 lines).
Hunk #4 succeeded at 639 (offset -9 lines).
patching file ui/mtr.h
patching file ui/net.c
Hunk #1 FAILED at 39.
Hunk #2 succeeded at 69 (offset -3 lines).
Hunk #3 succeeded at 85 (offset -3 lines).
Hunk #4 succeeded at 108 (offset -3 lines).
Hunk #5 succeeded at 176 (offset -3 lines).
Hunk #6 FAILED at 210.
Hunk #7 FAILED at 222.
Hunk #8 FAILED at 470.
Hunk #9 succeeded at 439 with fuzz 2 (offset -41 lines).
Hunk #10 succeeded at 497 (offset -43 lines).
Hunk #11 succeeded at 509 (offset -43 lines).
Hunk #12 succeeded at 527 (offset -43 lines).
Hunk #13 succeeded at 535 (offset -43 lines).
Hunk #14 FAILED at 642.
Hunk #15 succeeded at 590 (offset -98 lines).
Hunk #16 FAILED at 699.
Hunk #17 succeeded at 627 (offset -108 lines).
Hunk #18 FAILED at 755.
Hunk #19 succeeded at 687 (offset -113 lines).
Hunk #20 succeeded at 777 (offset -113 lines).
Hunk #21 succeeded at 799 (offset -113 lines).
7 out of 21 hunks FAILED -- saving rejects to file ui/net.c.rej
patching file ui/net.h
Hunk #1 succeeded at 115 (offset -2 lines).
patching file ui/raw.c
patching file ui/raw.h
patching file ui/report.c
Hunk #3 FAILED at 287.
1 out of 3 hunks FAILED -- saving rejects to file ui/report.c.rej
patching file ui/split.c
```

failures during the patch:

1.3.1

a) To list loaded kernel modules you use "lsmod" command.



```
kali@kali: ~  
Session Actions Edit View Help  
5:24 20K  
(kali@kali)-[~]  
$ lsmod  
Module              Size  Used by  
sctp                 540672  32  
ip6_udp_tunnel       16384  1 sctp  
udp_tunnel           36864  1 sctp  
udp_diag             12288  0  
tcp_diag             12288  0  
inet_diag            32768  2 tcp_diag,udp_diag  
ccm                  20480  6  
cmac                 12288  1  
nls_utf8             12288  15  
cifs                 1560576  3  
cifs_arc4            12288  1 cifs  
nls_ucs2_utils        8192  1 cifs  
cifs_md4             12288  1 cifs  
dns_resolver         12288  1 cifs  
netfs                606208  1 cifs  
snd_seq_dummy        12288  0  
snd_hrtimer          12288  1  
snd_seq              118784  7 snd_seq_dummy  
snd_seq_device       16384  1 snd_seq  
qrtr                 57344  2  
rfkill               45056  2  
intel_rapl_msr       20480  0  
vboxsf               49152  0  
intel_rapl_common    53248  1 intel_rapl_msr  
sunrpc               888832  1  
intel_uncore_frequency_common 16384  0  
intel_pmc_core       143360  0  
pmt_telemetry        16384  1 intel_pmc_core  
pmt_class            20480  1 pmt_telemetry  
intel_pmc_ssram_telemetry 16384  1 intel_pmc_core  
binfmt_misc          28672  1  
intel_vsec           20480  1 intel_pmc_ssram_telemetry  
rapl                 20480  0  
snd_intel8x0         57344  1  
snd_ac97_codec       212992  1 snd_intel8x0  
ac97_bus             12288  1 snd_ac97_codec
```

To load a kernel module you use the command "sudo modprobe " and if you want to unload a kernel module you can add "-r" like " sudo modprobe -r " or use the command "sudo rmmod "

```

kali@kali: ~
Session Actions Edit View Help
MODPROBE(8) modprobe MODPROBE(8)
NAME
modprobe - Add and remove modules from the Linux Kernel
SYNOPSIS
modprobe [-v] [-V] [-C config-file] [-n] [-i] [-q] [-b] [modulename]
[module parameters ...]
modprobe [-r] [-v] [-n] [-i] [modulename ...]
modprobe [-c]
modprobe [--dump-modversions] [filename]
DESCRIPTION
modprobe intelligently adds or removes a module from the Linux kernel: note that for convenience, there is no difference between _ and
- in module names (automatic underscore conversion is performed). modprobe looks in the module directory /usr/lib/`uname -r` for all
the modules and other files, except for the optional configuration files (see modprobe.d(5)). modprobe will also use module options
specified on the kernel command line in the form of <module>.<option> and blacklists in the form of modprobe.blacklist=<module>.
Note that unlike in 2.4 series Linux kernels (which are not supported by this tool) this version of modprobe does not do anything to
the module itself: the work of resolving symbols and understanding parameters is done inside the kernel. So module failure is some-
times accompanied by a kernel message: see dmesg(8).
modprobe expects an up-to-date modules.dep.bin file as generated by the corresponding depmod utility shipped along with modprobe (see
depmod(8)). This file lists what other modules each module needs (if any), and modprobe uses this to add or remove these dependencies
automatically.
If any arguments are given after the modulename, they are passed to the kernel (in addition to any options listed in the configuration
file).
When loading modules, modulename can also be a path to the module. If the path is relative, it must explicitly start with "./". Note
that this may fail when using a path to a module with dependencies not matching the installed depmod database.
OPTIONS
-a, --all
Insert all module names on the command line.
Manual page modprobe(8) line 1 (press h for help or q to quit)

```

b) When using the command "lsmod | grep fuse" nothing comes up. But you can check also if its active if "/dev/fuse" has presence:

```

kali@kali: ~
Session Actions Edit View Help
(kali@kali)~$ lsmod | grep fuse
(kali@kali)~$ ls -l /dev/fuse
crw-rw-rw- 1 root root 10, 229 Jan 20 11:05 /dev/fuse
(kali@kali)~$

```

1.3.2

a) For mounting i used "sudo sshfs -o allow_other,default_permissions,idmap=user,uid=(id -u),gid=(id -g) h23samum@maggie.du.se:/home/student/h23samum /mnt/remote_data". (remote_data is my mountingpoint) for unmounting i ran "fusermount -u /mnt/remote_data".

```
kali@kali: /
Session Actions Edit View Help

(kali@kali)-[~/mnt]
$ sudo sshfs -o allow_other,default_permissions,idmap=user,uid=$(id -u),gid=$(id -g) \
h23samum@maggie.du.se:/home/student/h23samum /mnt/remote_data
h23samum@maggie.du.se's password:

(kali@kali)-[~/mnt]
$ ls -ld /mnt/remote_data
drwx----- 1 kali kali 4096 Jan 23 06:03 /mnt/remote_data

(kali@kali)-[~/mnt]
$ cd /mnt/remote_data

(kali@kali)-[/mnt/remote_data]
$ ll
total 20
drwx----- 1 kali kali 4096 Mar 18 2024 Documents
drwx----- 1 kali kali 4096 Mar 18 2024 Downloads
drwx----- 1 kali kali 4096 Mar 18 2024 Favorites
drwx----- 1 kali kali 4096 Mar 18 2024 'Saved Games'
drwx----- 1 kali kali 4096 Mar 18 2024 Windesk

(kali@kali)-[/mnt/remote_data]
$ cd Saved\ Games

(kali@kali)-[/mnt/remote_data/Saved Games]
$ ll
total 8
drwx----- 1 kali kali 4096 Mar 18 2024 '$RECYCLE.BIN'
-rwx----- 1 kali kali 282 Mar 18 2024 desktop.ini
```

b) Advantages:

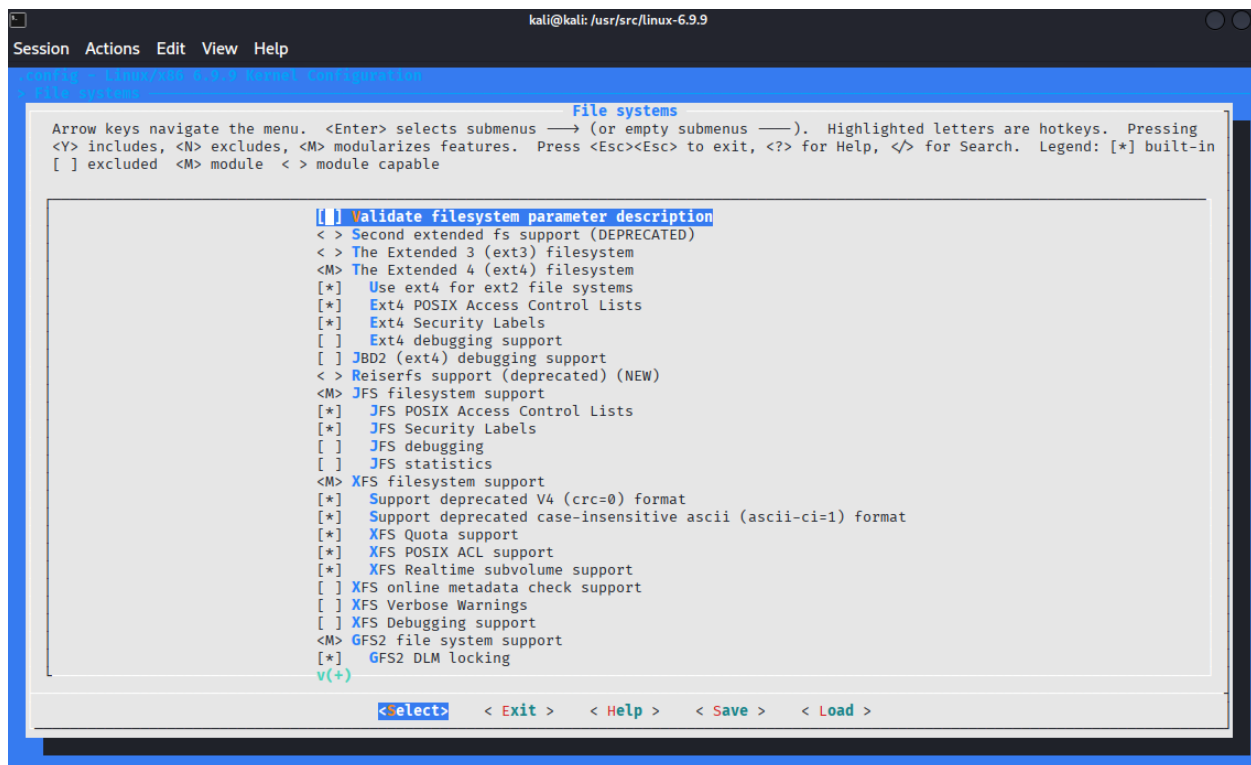
- SSHFS is encrypted by default via ssh.
- Easy to set up and remove
- Works over LAN and WAN

Disadvantages:

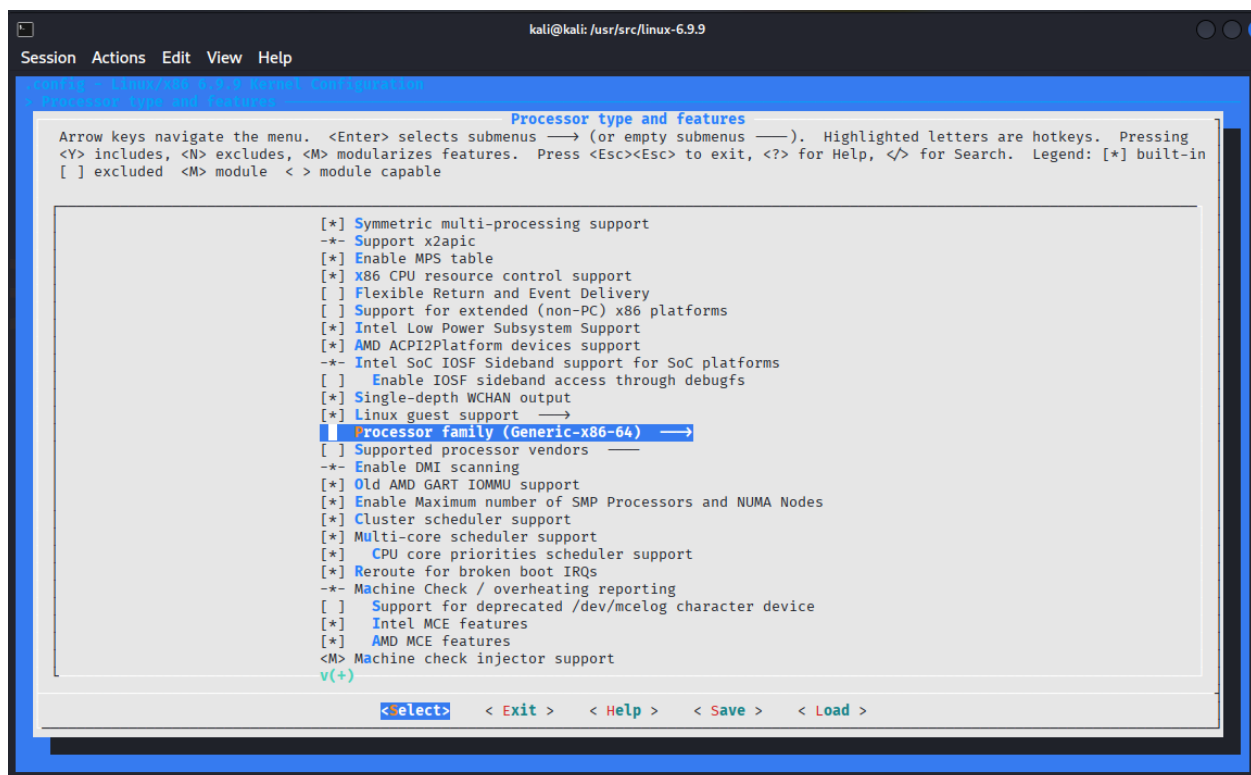
- Slower than NFS or local filesystems
- Sensitive to latency
- Connection drops and breaks the mount.

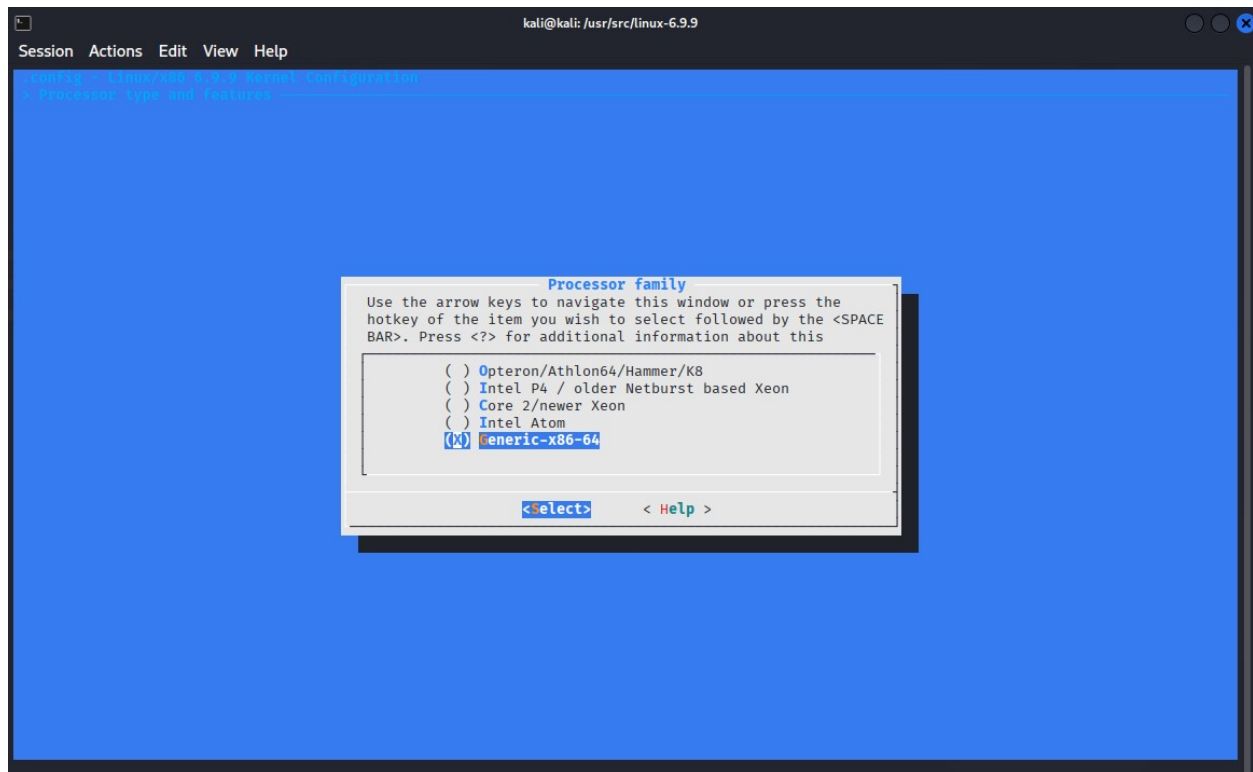
1.3.3

a)



b)





c)

- "make menuconfig" configure kernel features.
- "make" command compile the kernel and modules.
- "make modules_install" installs kernel modules
- "make install" installs kernel image
- "update-grub" registers a new kernel.
- "reboot" boots into the kernel.

1.4.1

a) I used the command "systemctl status ssh". Here we can see that the daemon is not active(running)

```

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[/usr/src/linux-6.9.9]
$ sudo -i
(root@kali)-[~]
# systemctl status ssh
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)

```

b) For starting the daemon I used "systemctl start ssh". For stopping it I used "systemctl stop ssh"

```
root@kali: ~
Session Actions Edit View Help

(root@kali)-[~]
# systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-01-21 10:51:44 EST; 3s ago
  Invocation: 428d2ef669b146c0952176fa3e72f29e
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 86625 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 86628 (sshd)
    Tasks: 1 (limit: 2116)
  Memory: 1.9M (peak: 2.6M)
     CPU: 65ms
  CGroup: /system.slice/ssh.service
          └─86628 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 21 10:51:43 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Jan 21 10:51:44 kali sshd[86628]: Server listening on 0.0.0.0 port 22.
Jan 21 10:51:44 kali sshd[86628]: Server listening on :: port 22.
Jan 21 10:51:44 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

```
root@kali: ~
Session Actions Edit View Help

CPU: 65ms
CGroup: /system.slice/ssh.service
        └─86628 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 21 10:51:43 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Jan 21 10:51:44 kali sshd[86628]: Server listening on 0.0.0.0 port 22.
Jan 21 10:51:44 kali sshd[86628]: Server listening on :: port 22.
Jan 21 10:51:44 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(root@kali)-[~]
# systemctl stop ssh

(root@kali)-[~]
# systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead) since Wed 2026-01-21 10:57:12 EST; 5s ago
  Duration: 5min 28.839s
  Invocation: 428d2ef669b146c0952176fa3e72f29e
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 86625 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Process: 86628 ExecStart=/usr/sbin/sshd -D $SSH_OPTS (code=exited, status=0/SUCCESS)
 Main PID: 86628 (code=exited, status=0/SUCCESS)
  Mem peak: 2.6M
     CPU: 67ms

Jan 21 10:51:43 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Jan 21 10:51:44 kali sshd[86628]: Server listening on 0.0.0.0 port 22.
Jan 21 10:51:44 kali sshd[86628]: Server listening on :: port 22.
Jan 21 10:51:44 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jan 21 10:57:12 kali systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server ...
Jan 21 10:57:12 kali sshd[86628]: Received signal 15; terminating.
Jan 21 10:57:12 kali systemd[1]: ssh.service: Deactivated successfully.
Jan 21 10:57:12 kali systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
```

1.4.2

The image only shows when i tried diffrent commabs and key combinations.

1. Create a tmux session - "tmux new -s "
2. To detach from session press CTRL + b then d
3. To list active tmux sessions use command "tmux ls"
4. To reattach "tmux attach -t "

5. To kill the session use "tmux kill-session -t

```
Session Actions Edit View Help
(kali@kali)-[~]
$ tmux new -s work
[detached (from session work)]

(kali@kali)-[~]
$ tmux ls
work: 1 windows (created Wed Jan 21 11:03:25 2026)

(kali@kali)-[~]
$ tmux attach -t work
[exited]

(kali@kali)-[~]
$
```

1.5

1.5.1

a) I used the command "nc -lvp 4444" to start listening on port 4444 and for the client (other terminal) makes it listening to itself.

```
Session Actions Edit View Help
(kali@kali)-[~]
$ nc -lvp 4444
listening on [any] 4444 ...
connect to [127.0.0.1] from (UNKNOWN) [127.0.0.1] 44994
hi

```

```
Session Actions Edit View Help
(kali@kali)-[~]
$ nc 127.0.0.1 4444
hi

```

1.6 Feedback

a) I like the lab and informed about some nice tools, some of it i have done before which made me feel comfortable with the lab.

b) Nothing to it, it gives a simple understanding of what you can do with linux and kali.